Data-driven recommendations for increasing reliability in measures of infant word recognition

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Abstract

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Introduction

Over the first few months of life, children become experts on their native language, organizing the phonological system of the speech that they hear and parsing speech into meaningful units. A bit later, children begin to show early signs of comprehension of words or phrases through simple behaviors such as turning their head when their name is called (e.g., "Hey, Billy!") or reaching for an object that is named (e.g., "Where's the ball?"). These everyday accomplishments mark important milestones in children's development, yet such global assessments may miss continuous processes of growth in underlying skills and the millisecond-level processes that take place during real-time language comprehension. Children, like adults, process speech incrementally, as it unfolds in time (Fernald, Pinto, Swingley, Weinberg, & McRoberts, 1998; Frank et al., 2025; Snedeker, 2013), and use features of the speech signal to extract meaning that will support language comprehension (Weisleder & Fernald, 2013; **law2015?**).

These moment-to-moment processes are difficult to document without child-friendly procedures that allow precision in capturing how children's behaviors are time-locked to speech. In particular, the "looking while listening" (LWL) procedure has emerged as a *de facto* standard for measuring language in infants and children, especially when they are younger than two years old, as other options for direct assessment during this period are limited (Frank, Braginsky, Yurovsky, & Marchman, 2021).[1] In the standard version of LWL, children see two pictures on a screen and hear a referring expression, e.g. "look at the ball"; their time-locked eye-movements to the ball over the distractor image are then taken to be evidence for language comprehension (Fernald et al., 2008).

LWL has been a key tool for enhancing our understanding of infants' language processing and knowledge. For example, using LWL methods, we can measure increases in speed and accuracy in responding to familiar words (Fernald, Perfors, & Marchman, 2006; Frank et al., 2025) and increases in the detail of their phonological representations (Moore & Bergelson, 2024; Swingley & Aslin, 2000) during the first years after birth. Moreover, measures of children's skill in the LWL task have also proven useful as an index of individual differences. For example, children who show more efficient language processing at 18 months show larger vocabulary size and more rapid vocabulary growth over the second and third years than children with less efficient processing (Fernald & Marchman, 2012; Fernald et al., 2006; Frank et al., 2025; Peter et al., 2019). Language processing efficiency has also been shown to link to later language, cognitive, and non-verbal outcomes in both typically-developing (Peter et al., 2019; **law2014?**) and preterm children (Virginia A. Marchman, Adams, Loi, Fernald, & Feldman, 2016; V. A. Marchman, Dale, & Fenson, 2023; Virginia A. Marchman et al., 2018).

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[1] Here we adopt LWL as a convenient label while recognizing that the procedure goes by a number of different names in the literature including the "intermodal preferential looking procedure" (Fernald, Zangl, Portillo, & Marchman, 2008; Reznick, 1990; **hirsh1996intermodal?**).

One key reason for the success of this research is the care that has been invested in the development of the looking-while-listening method. Past work has articulated best practices in collecting and analyzing looking-while-listening measures, with a particular focus on stimulus and design choices that make looking preferences maximally informative about word recognition processes (Fernald et al., 2008). However, despite widespread use of the looking-while-listening method in language development research, there is still tremendous variation and limited consensus on how to best process and analyze time-series data of this type [from either adults or infants; (**mirman2008?**);Bergelson and Swingley (2012);Peelle and Van Engen (2021);Von Holzen and Mani (2012)].

This lack of consensus likely stems from structural factors: infant recruitment is time- and cost-intensive and datasets collected by individual labs are often small (Bergmann et al., 2018; Frank, Braginsky, Yurovsky, & Marchman, 2017). These issues make it difficult for any one researcher to systematically evaluate the consequences of specific analytic choices independently of the data collected for a particular study (cf. Peelle & Van Engen, 2021). Instead, researchers are often forced to make ad-hoc decisions or to adopt idiosyncratic lab practices that themselves originate from ad-hoc subjective judgments of what "seems to work". How can we put the measurement and analysis of infant word recognition and online language processing on firmer empirical footing?

Peekbank is a new data resource that offers a potential solution to this core issue by providing a large database of infant eye-tracking data from the looking-while-listening paradigm for exploring potential methodological questions (Zettersten et al., 2022). The most recent release of Peekbank at time of writing (version 2026.1), currently houses 44 datasets from infants across a wide age range [6 months - 5 years] and across a broad range of test words and materials, resulting in observations from over 3915 infants participating in the looking-while-listening task. The goal of the current manuscript is to use these data to systematically explore key analytic decisions that researchers make when using this paradigm, offering best-practices guidance for measurement.

Our approach throughout is to seek analytic choices that distill the best possible measures from the rich time course of behavior in the looking-while-listening task. Ideally, these measures should be reliable, yielding the maximal signal about individual children relative to both measurement error and irrelevant artifacts (Byers-Heinlein, Bergmann, & Savalei, 2022; Rhemtulla & Bottesini, 2022; Schreiner et al., 2024). Here, to measure reliability, we use both within-dataset inter-item reliability (ICC) and test-retest reliability within datasets with multiple LWL sessions close in time for the same children.

Further, LWL measures should be valid, meaning operationally that they should relate to other measures that are thought to reflect related constructs (Flake & Fried, 2020; Kominsky, Lucca, Thomas, Frank, & Hamlin, 2022). Here, we select as our primary measure the correlations between LWL measures and parent reports of vocabulary measured with the MacArthur-Bates Communicative Development Inventories [MB-CDI, a popular parent report measure of early vocabulary that is widely used in conjunction with LWL tasks; Frank et al. (2021); Fenson et al. (1994)]. Here we are specifically interested in the relationships

between aggregate measures of vocabulary size from the MB-CDI and LWL data, given both the documented relationships between these (reviewed in V. A. Marchman et al., 2023) and the mixed evidence on item-level alignment between LWL and MB-CDI data (López Pérez, Moore, Sander-Montant, & Byers-Heinlein, 2024). This correspondence reflects the general relationship between word recognition in the moment (indexed by LWL) and the growth of early vocabulary over time (indexed by the MB-CDI) (Frank et al., 2025).

Our focus here is on providing guidance for navigating some of the many decisions facing an investigator analyzing LWL data. ((**sanderkeeping?**) provide a similar analysis, also using Peekbank and focusing on the selection of different dependent measures and the general construct structure of LWL measures). One common measure of LWL is *accuracy*: how much of the time a child looks at the target image compared to looking at either the target or distractor image over some period of time after the auditory stimulus. Children's responses unfold over time, but it is not known what start and end time maximizes the signal of children's response to the stimulus (Peelle & Van Engen, 2021). A second commonly used measure is *reaction time* – how long it takes a child to change where they are looking in response to the stimulus (Fernald et al., 2008). There are even more analytic options available for this metric – whether to measure to a child leaving the distractor or looking at the target, whether to measure RT on a log or raw scale, and how long after the start of the stimulus to start counting the RTs. For both of these measures, there are also questions about whether and when to exclude either children or trials.

While the LWL task is used for a wide variety of purposes, including measurements of individual differences (e.g., Peter et al., 2019) as well as measurements of experimentally-induced condition differences (e.g., Swingley & Aslin, 2000), here we focus only on what we refer to as "vanilla" word recognition items. These are items that test vocabulary directly with familiar targets, familiar distractors, and no informative carrier phrases or experimental manipulations. We focus on these items both because they are the most commonly used LWL items and because they are the case in which reliability and validity is most clearly defined. Vanilla trials are expected to reflect the construct of word knowledge and correlate with other measures of word knowledge, including vocabulary size.

The plan of the paper is as follows. We begin by overviewing the Peekbank data. We then make a set of targeted recommendations, providing evidence for each based on the impact of our recommendation on the average reliability and validity of the resulting measures. To foreshadow our results, we find that many common data analysis practices provide relatively reasonable levels of reliability and validity, but it may be possible to optimize reliability and validity through careful decision-making.

Methods: Dataset and approach

Peekbank is an aggregation of datasets from a number of LWL studies. This work is based on Peekbank version 2026.1 (released February 2026) which has 134,929 trials from 3915 children and 44 datasets.

Inclusion criteria

We focus our exploration of data processing choices on a set of items where we have the clearest expectations around what we are trying to measure. While LWL is also used to study a variety of experimental effects, such as the effects of noise, novel word learning, expectations of mutual exclusivity, inferences from selective verbs, blah blah, these items may index different constructs. We limit our analyses to trials where both the target and distractor are real nouns, the carrier phrase is neutral with no experimental manipulation. TODO list out full vanilla criteria While still heterogenous, these "vanilla" trials have a shared interpretation for how looking behavior is expected to relate to underlying language knowledge and processing. We only include trials from childen 60 months or younger. We do not place restrictions on the language of the trials or children's language background.

Descriptives

This paper is based on 36 datasets, comprising 3171 children, 4851 administrations, and 75636 trials. There is MB-CDI comprehension scores for 632 children from 12 of the datasets, and MB-CDI production scores for 1143 children from 20 of the datasets. Our test-retest data comes from 5 datasets with multiple administrations within 1.5 months of each other, covering 365 children (788 administration pairs). See Table 1 for a list of included datasets.

Table 1
Descriptives of the included datasets.

	Dataset Name	Pct Trials	Trials	Admins	Subjects	Age	CDI	Test-retest	Language
1	Adams et al. (2018)	18.4%	13899	711	69	22 [13 - 38]	x	x	English
2	Fernald & Marchman (2012)	15.6%	11828	679	122	22 [17 - 32]	x	x	English
3	Weaver et al. (2024)	6.5%	4948	248	141	16 [14 - 24]		x	English
4	Fernald et al. (2013)	5.8%	4359	178	80	20 [17 - 26]	x	x	English
5	Hurtado et al. (2008)	5.3%	4035	136	76	21 [17 - 27]	x		Spanish
6	Fernald et al. (2006)	4.9%	3692	229	63	20 [15 - 25]	x		English
7	Yurovsky et al. (2013)	3.7%	2810	413	413	34 [12 - 60]			English
8	Yurovsky et al. (2017)	3.1%	2350	327	327	37 [8 - 60]			English
9	Bergelson & Swingley (2012)	2.8%	2081	84	84	12 [6 - 21]	x		English
10	Kartushina & Mayor (2019)	2.7%	2072	69	69	7 [6 - 9]	x		Norwegian
11	Weaver & Saffran (2026)	2.4%	1788	69	69	19 [18 - 20]	x		English
12	Weisleder & Fernald (2013)	2.3%	1769	58	29	22 [18 - 27]			Spanish
13	Yurovsky & Frank (2017)	2.3%	1747	285	285	26 [13 - 59]			English
14	Borovsky & Peters (2019)	2.2%	1676	79	79	18 [17 - 20]	x		English
15	Potter & Lew Williams (2023)	2.1%	1626	67	67	24 [21 - 27]	x		English
16	Yoon et al. (2015)	2.0%	1500	199	199	43 [13 - 60]			English
17	Sander-Montant et al. (2023)	1.8%	1381	123	123	22 [12 - 31]	x		English, French
18	Mahr et al. (2015)	1.6%	1232	29	29	21 [18 - 24]	x		English
19	Egger et al. (2020)	1.4%	1052	54	54	18 [18 - 19]	x		Dutch
20	Garrison et al. (2020)	1.4%	1046	35	35	15 [12 - 18]	x		English
21	Hurtado et al. (2007)	1.3%	1010	49	49	24 [15 - 37]			Spanish
22	Pomper & Saffran (2017)	1.0%	785	82	82	17 [14 - 19]			English
23	Frank et al. (2016)	1.0%	781	108	108	35 [12 - 60]			English
24	Bacon & Saffran (2022)	1.0%	754	38	38	23 [22 - 24]	x		English
25	Ronfard et al. (2022)	1.0%	737	40	40	20 [18 - 24]	x		English
26	Perry & Saffan (2017)	0.9%	718	42	42	20 [19 - 22]	x		English
27	Casillas et al. (2017)	0.9%	651	23	23	32 [9 - 48]			Tseltal
28	Swingley & Aslin (2002)	0.8%	596	50	50	15 [14 - 16]	x		English
29	Pomper & Saffran (2016)	0.6%	471	60	60	44 [41 - 47]			English
30	Kremin et al. (2023)	0.5%	410	60	60	42 [37 - 49]			English, French, Spanish
31	Pomper & Saffran (2015)	0.5%	389	25	25	40 [38 - 42]			English
32	Moore & Bergelson (2022)	0.5%	384	29	29	18 [16 - 20]	x		English
33	Potter et al. (2019)	0.4%	309	44	23	23 [18 - 29]	x	x	English
34	Byers-Heinlein et al. (2017)	0.4%	284	48	48	20 [19 - 21]	x		English, French
35	Pomper & Saffran (2019)	0.3%	249	44	44	40 [38 - 43]			English
36	Pomper & Saffran (2018)	0.3%	217	37	37	39 [38 - 43]			English
	Total	100%	75636	4851	3171	22 [6 - 60]	21	5	

Approach

Throughout we use two reliabilty metrics and two validity metrics.

An intra-class correlation (ICC) as a measure of reliability. Specifically, we use a measure of inter-rater reliability that measures (within a dataset) how much different items (target labels) are consistent with each other in "rating" the subjects (kid-administrations).

Test-retest reliability for measures on the same child at two different administrations within 1.5 months of each other. This measure is only possible on a subset of the datasets that closely spaced repeat administrations. While we are limited in how much data we have for test-retest reliability, this is the metric that is most important for measuring stable individual differences.

For validity, we correlate with CDI production measures and CDI comprehension measures, from the datasets where we have CDI administrations. We use both short and long form CDIs, using the fraction of items a child was marked as knowing as our measure. In general, correlating with production and comprehension in the expected directions is desirable, although we also may expect that LWL measures are picking up on slightly different aspects of language learning than is represented in CDI.

Additional details about how these were calculated are in Supplement XX.

All our code etc etc is available at somewhere.

Data processing recommendations

Looking while listening data takes a lot of processing to go from the experiment to the data that is analysed. Experiment design, and eye-tracking and eye-gaze coding considerations are covered elsewhere [citations], so we focus here on the process of going from a series of timepoints to the measures of accuracy and reaction time that are analysed. There are a number of decisions involved, including what windows and time cutoffs to use, and what exclusions to make around missing or low amounts of data.

Recommendation 1: Measure accuracy for more than 2 seconds post disambiguation

In LWL studies, accuracy is typically calculated as the fraction of time a child is looking at the target divided by the time they are looking at the target or the distractor. These fractions are computed over some window of time, but the selection of this window is complex. Typically windows start shortly after the point of disambiguation and continue until several seconds later; this window is intended to capture the peak of children's stimulus-driven looks to the target. However, in practice studies vary in what windows are used, and often window selection is a locus for analytic flexibility (choosing windows post hoc to maximize a particular effect) (Peelle & Van Engen, 2021). In the literature, it is common to use windows from 300ms – 1800 ms (following Fernald et al., 2008) or from 367ms -2000 ms(following **swingleyaslin2022?**). Windows that last for longer are less

commonly used – example until 4000ms in (**bergselsonswingley2017semantic?**) 3500ms in (**bergelsonswingley2012?**) or (**kartushina?**) or until 2800ms in (**weaverzettersten2024?**) or 3000 ms (**lopez_perez_2025?**).

Some analysts recommend relatively short windows that start several hundred milliseconds after stimulus onset and end around 1500 milliseconds later (e.g., Fernald et al., 2008), however practices vary throughout the literature. In this analysis, we begin from first principles and ask what windows maximize reliability and validity on word recognition trials.

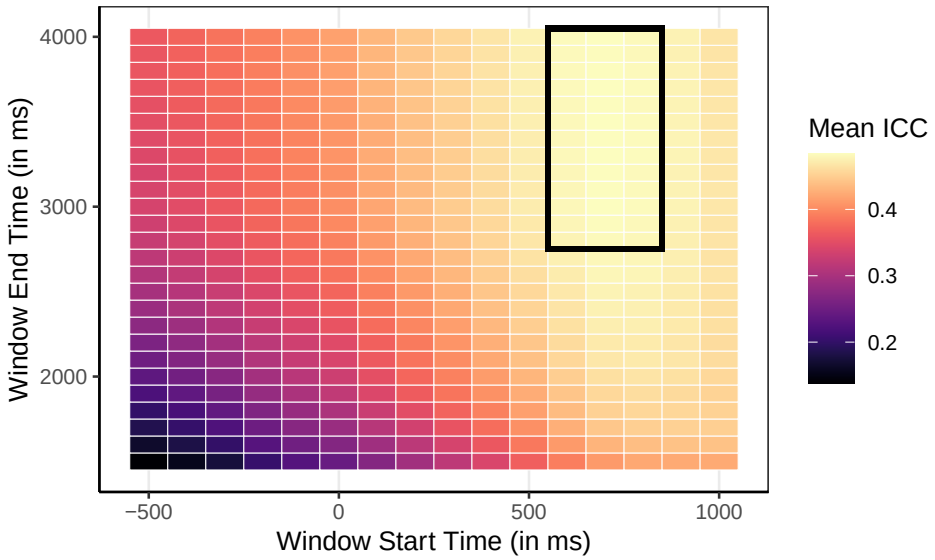


Figure 1. A heat map showing the average ICC across datasets for different start and end points for the accuracy window. A region with the highest ICC (best reliability) is outlined.

Analytic approach. We first calculated the ICC reliability for accuracy across datasets, computing average ICC across a range of different window onsets and offsets. Next, to further explore the impact of window size, we focused on the comparison between windows ending at 2000ms, 3000ms, and 4000ms as well as fine-grained options for window onset between 0 and 1000ms. ICC, CDI correlations, and test-retest reliability were all calculated by dataset, for each combination of parameter values. To aggregate across datasets, we used a linear mixed effect model with random intercept by dataset, weighted by the number of administrations. We interpret the fixed-effect contrasts as what differences in outcome measure would be expected from switching from one set of parameters to another on a generic dataset. We report estimates and 95% confidence intervals. [refer to Supplement YY for details].

Because we were concerned that the best data-processing options could be different for children of different ages, we ran the same linear mixed effect models separately for data just on children in each of 4 age groups: <18 months, 18-24 months, 24-36 months, and >36 months.

Results. Average ICCs across windows are shown in Figure 1. As expected, including time points from before the point of disambiguation decreases reliability. Our initial exploration found a region of high reliability with onsets in the 600-800 range and

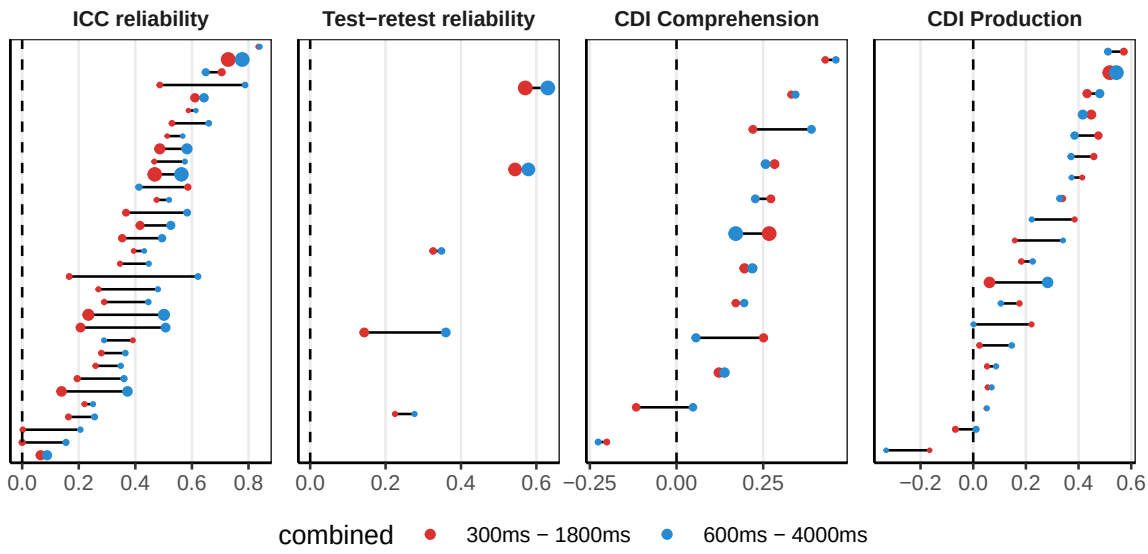


Figure 2. Per dataset outcome measures for two window options: 300ms-1800ms (red) and 600ms-4000ms (blue). Dot size is proportional within each panel to the number of administrations.

offsets in the 2800-4000 range, surrounded by a gradual decline in ICC reliability.

A dataset by dataset comparison of the differences between a 200-2000ms window and a 600-4000 ms window are shown in Figure 2. Nearly all datasets have higher ICC reliability for the later and longer window than the shorter window that is more typical of current practice. All 5 datasets with close in time pairs of administrations also show increased reliability on this later and longer window than the shorter window. Correlations with CDI vary across datasets, with an overall similar effect for both windows.

Our mixed effect model confirms that there are expected reliability gains from switching from a 300-1800ms window to a 600-4000ms one. ICC reliability is expected to increase by 0.12 [0.08, 0.16] and test-retest reliability by 0.07 [-0.02, 0.16]. There are limited differences in validity between these two windows (-0.02 [-0.07, 0.03] for comprehension, and 0.03 [-0.02, 0.08] for production).

There are a wide range of window options with generally comparable reliability and validity measures (Fig S3 in supplement). For instance, a 400 - 3000 ms window is not significantly different from a 600 - 4000 ms window on any of the outcome measures.

While the looking signal between 2 and 4 seconds is weaker than between .5 and 2 seconds, there is still signal, especially just after the 2 second mark (even for older children, see Figure SNN).

Exactly when the accuracy window starts is not important, although we recommend starting between 400ms and 600 milliseconds to optimally balance validity and reliability. Continuing for a long window also increases reliability.

These patterns generally hold across age. The overall level of reliability rises with age, as younger children have less signal, and experiments with younger children may also use fewer trials (see Figure S2 for an age-split version of the ICC heat map). However, to make data-driven recommendations, a question is whether there are substantive differences in what windows maximize validity and reliability across age.

While the strength of evidence varied, the overall pattern of results was remarkable consistent across the different age groups (see Figure S4). For instance, across different ages there is an improvement in ICC for using a 600-4000 window compared to a 300-1800ms one. For <18 months, the ICC difference is 0.06 [-0.03, 0.16]; for 18-24 months, 0.13 [0.07, 0.19]; for 24-36 months, 0.06 [0.01, 0.12]; and for 36+ months 0.15 [0.06, 0.23]. The main difference that is seen across ages is that when there is less signal to begin with, there's less difference between different measurements.

For CDI production, there might be slight age effects, but the differences are not significant and are difficult to interpret. For <18 months, a longer window results in change of 0.02 [-0.07, 0.12] in correlation with CDI production ; for 18-24 months, -0.03 [-0.08, 0.02]; for 24-36 months, 0.06 [-0.03, 0.15]. These small and inconsistent age effects could be due to dataset differences across different ages or small but true developmental differences.

Strong effects are likely to show up in any reasonable window, but for eking the most reliability out of data, we recommend windows that start somewhere in the 400-600 ms range and go for longer than 2000 ms. Why would later starting and later ending windows increase reliability? This suggests that there is some signal in the post 2000ms window, and that there may not be so much signal in the early 200-400 ms time. While some children may look quickly to the target and then look away, other children who look to the target later or who continue looking at the target at still contributing signal in this region, and overall children in many datasets are in aggregate looking to the target more than the distractor more in this region (even if less than at the peak, Figure SZZ). Importantly, since accuracy is measured as the ratio of target looks to either target or distractor looks, looking completely away does not count against accuracy.

Recommendation 2: Do not use baseline-corrected accuracy

One commonly acknowledged issue in LWL studies is that looking behavior could be partly driven by preferences over the target images themselves (Fernald et al., 2008; **novack2025development?**): Children may have a baseline preference to look at one image over another, even in the absence of auditory stimuli. Although thoughtful design choices can minimize these issues (see Recommendation 6), some researchers also attempt to deal with baseline differences in stimulus salience in their analysis. This practice is known as "baseline correction": at the data-processing stage, researchers use a difference measure between accuracy on a post-stimulus window and "accuracy" (looks to target versus looks to target or distractor) on a pre-stimulus window (Bergelson & Swingley, 2012; Weaver, Zettersten, & Saffran, 2024). Pre-stimulus windows are calculated on the 2seconds before POD (weaver x 2) or for the entire pre-stimulus time 3-4 seconds ((**bergelsonswingley2012?**))

Analytic approach. We considered baseline-corrected accuracies with various starting and ending points for the baseline window, and used a 600ms-4000ms main window (although the results generalize to other main windows). We restricted ourselves to datasets with at least some pre-0 data. We also included a "no baseline" version which was just accuracy on 600-4000ms with no correction. Baseline windows started at -1000, -2000, -3000 and -4000ms and ended at -500 or 0 ms.

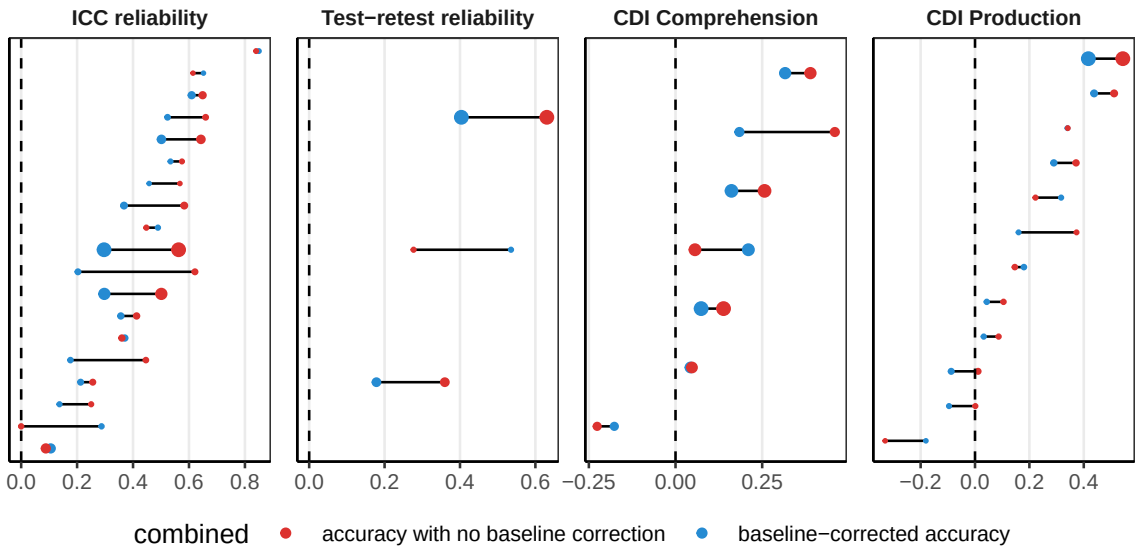


Figure 3. Comparison between a baseline-corrected accuracy in red and accuracy in blue across datasets and measures. Both use an accuracy window of 600 - 4000ms, the baseline window was -2000 - 0ms.

Results. Baseline-corrected accuracy led to worse reliability and validity than accuracy.

As an example, Figure 3 shows per-dataset results for accuracy compared to a baseline-corrected accuracy using a -2000 - 0 baseline window. All baseline correction options that we considered led to significantly worse reliability and validity measures compared to using normal accuracy without any correction. For a regular window of 600-4000, baseline correction is expected to make ICC worse by at least 0.1 [0.04, 0.16]; test-retest worse by at least 0.19 [0.08, 0.3]; correlation with comprehension worse by at least 0.04 [-0.05, 0.12] and correlation with production worse by at least 0.09 [0.05, 0.13].

Similar patterns hold for different accuracy windows (Fig SXX maybe) and across different age bins (Fig SYY). Even restricting to the datasets with the longest pre-windows, there are still not benefits to baseline correction (Figure SXX).

Why would incorporating baseline information lead to worse metrics? There is a complex literature on the psychometrics of difference scores and when they result in lowered reliability (e.g., **lord1956measurement?**; **rogosa1983demonstrating?**; **tramifow2015defense?**). The situation for looking while listening data is that children's looking behavior is noisy but the signal across baseline and critical portions of a trial is likely correlated (in part due to the very salience that the baseline correction is trying to remove).

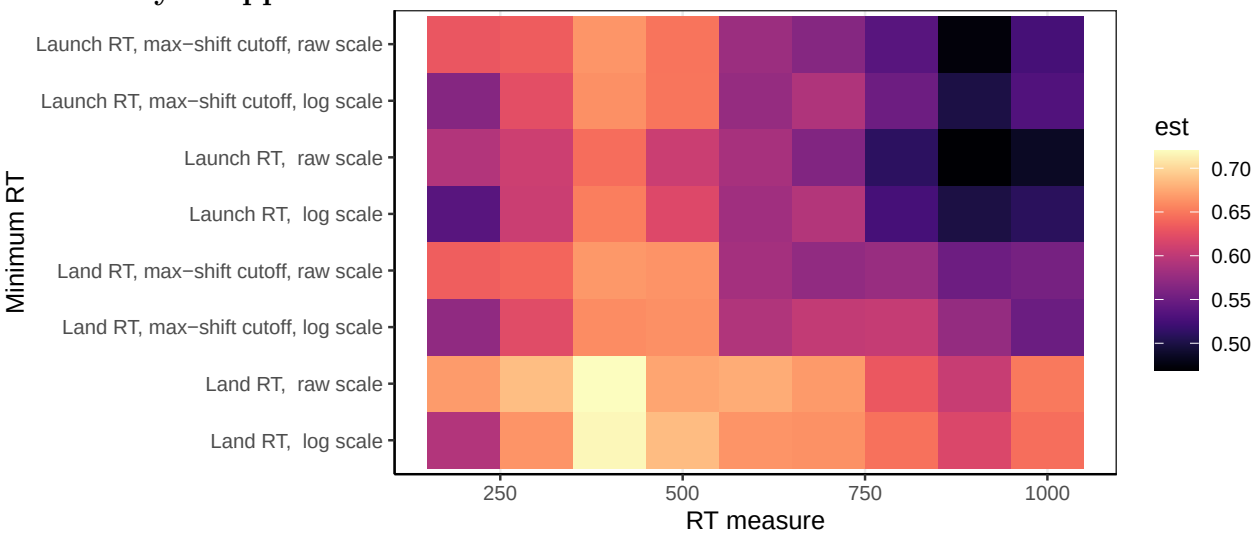
Unfortunately, it is under precisely these conditions – low reliability and high correlation between components – that difference scores exhibit the lowest reliability (**trafimow2015defense?**). Thus, while baseline-correction may lead to less biased grand average accuracy scores (by removing item salience), it also leads to less individual-level measures.

Recommendation 3: Measure reaction times with a minimum reaction time of around 400 ms

In addition to accuracy measures, LWL studies also provide the potential for computing children's reaction time [RT; Fernald et al. (2008)]. RT is typically computed on trials when the child is initially looking at the distractor and shifts to the target after stimulus onset, as there is not a way to measure how quickly a child decides to keep looking at the target. RT computations include many choice points:

commonly filtered to 300-1800 (following fernald2008) - **Launch vs. landing**: RTs can be computed based on when a child's eyes leave from the distractor or when a child's eyes land on the target. - **Exclusion of short RTs**: Short RTs are often excluded because they are considered unlikely to be generated in response to the stimulus. - **Linear vs. logarithmic analysis**: Reaction time data are often analyzed using a logarithmic transform (consistent with the idea that reaction times are log-normally distributed and that effects on them may be multiplicative rather than additive) (**csibra2016statistical?**; **frank2025continuous?**).

Analytic approach.



In our initial exploration, we considered a wide range of options for measuring RT, fulling crossing a) launch or land based, b) removing RTs with a shift-length greater than 600 ms (or not), c) using a log or raw scale, and d) minimum RTs between 0 and 1000 ms. (note: fuller range still needs to run)

For the main validity and reliability measures, we compare a launch-based RT with a 600ms shift-length cutoff (as used in . . . TODO) and a land-based RT without a shift-length cutoff (as used in . . . TODO). We vary the minimum RT and whether the RTs are on a log or

raw scale.

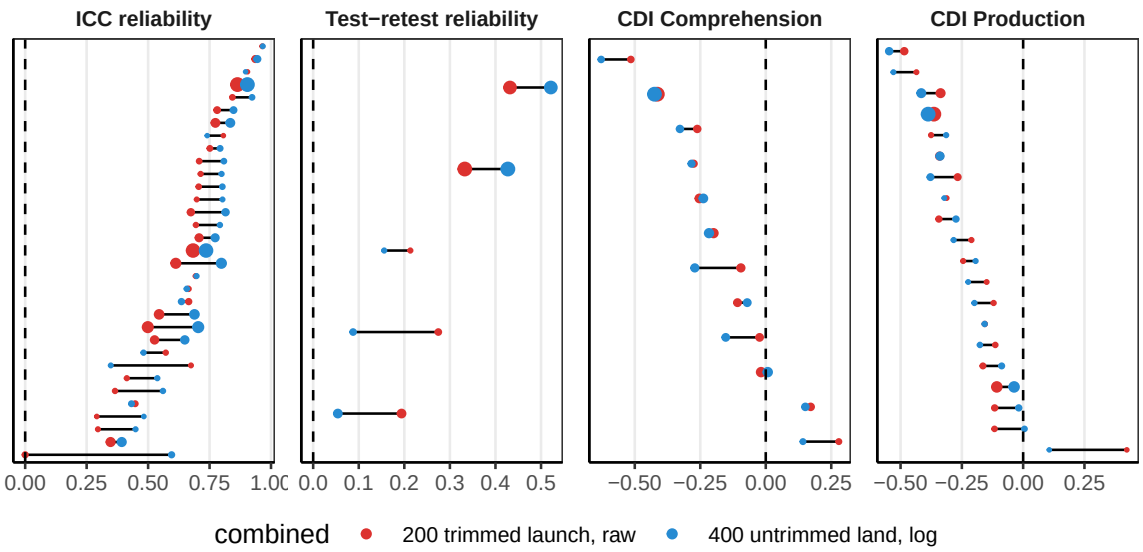


Figure 4. Dataset differences in RT measurement

Results. An initial exploration of ICC reliability (Figure **??**) indicated that ICC reliability was maximized for log-scale landing-based RT without any shift-length cutoff with minimum RTs around 400 ms. For launch-based RT, trimming for only shifts that landed within 600ms of launch had higher ICC than when they were not trimmed (and this is the common practice for launch based CITATION).

A comparison between two different ways of calculating RT is shown in Fig 4.

Compared to a minimum of 200 ms launch-based RT raw-scale, a land-based minimum of 400 RT (logged) is expected to increase ICC by 0.09 [0.01, 0.17] and possibly increase test-retest reliability by 0.04 [-0.11, 0.2]. Differences in CDI correlation are not substantive (0.03 [-0.05, 0.12] for comprehension, and 0.01 [-0.05, 0.07] for production).

These patterns generally hold for children of different ages (Figure SXX).

We recommend that researchers use minimum RT cutoffs around 400 ms as these increase reliability compared to shorter cutoffs like 200 ms. While the differences between different methods are small, we recommend using landing-based RT on the log scale to maximize overall reliability.

Overall RT measures complement accuracy measures and capture substantial signal; the exact details don't matter that much.

Recommendation 4: Trial-level and child-level exclusions for low amounts of data are generally unnecessary.

[why people exclude, range of practice] A common issue with developmental research is when to exclude trials for behavioral patterns that don't seem to match the task and when to exclude children for contributing too few valid trials.

For LWL studies, trial-level exclusions tend to be either for contributing too few time points (ex. too much tracker loss or a child looking away from the screen too much) [citations] or for "zoning" where a child only looks at one image and not the other. Zoning exclusions are sometimes done for trials where one image was not fixated at all in the entire trial, but can also be done for trials where one image was not fixated prior to the start of the accuracy window.

If trials with zoning or with few valid looking points still provide signal of the child-level language behavior, excluding them will decrease reliability and validity. On the other side, if these trials are not representing the construct and are just noise, excluding them will increase reliability and validity.

trial level exclusions: * pomper saffran is clearest for 2/3 on target requirement * some do 500ms continuous instead * or generic inattentive * or .5 weaver zettersten

- some require looking to T and/or D at t=0
- kid level – often requirement for up to half of trials (varies by unit)
- but for RT often does as min # of RTs – usually 2 (cf various marchman papers all do this), but (distinctly unclear if accuracy/kid metrics also affect RT or not)

Analytic approach

For accuracy measures, we consider both trial-level and administration-level exclusions. We first explored trial-level exclusions, both on the basis of amount of missing or away looks and on the basis of when both target and distractor were looked at (Figure **??**).

We tested exclusions for what portion of the time

TODO check what the window was for each of these and possibly adjust to match the window we're using!

Results

Compared to not setting any limits on the amount of missing data or looking away, a requirement to look at the target and distractor at least 50% of the time results in 85.64 % (range:27.22% - 98.92%) trials being retained. These exclusions lead to an expected increase in ICC by 0.03 [-0.07, 0.12] and increase in test-retest reliability by -0.02 [-0.17, 0.13]. Differences in CDI correlation are not substantive (0.01 [-0.11, 0.12] for comprehension, and 0 [-0.07, 0.07] for production). CHECK TEXT

TODO what is the actual range of practice with regard to exclusion rates and types of zoner exclusions? On a trial by trial basis, ICC tends to slightly increase the more noisy trials are excluded, but this comes at the cost of fewer trials. CDI correlations are higher when there are no zoner exclusions; test-retest is highest if there are no zoner exclusions or if exclusions are only done for the whole window (not just pre-window). Overall, based on

these different trade-offs, we recommend making light trial-level exclusions if any. It is reasonable to not filter trials at all and include those with any looking, but is also to set a minimum proportion data for up to 50%. More stringent looking requirements will mostly reduce the amount of useable data without validity gains to justify it.

We then explored the effects of excluding children for too few trials. Because datasets vary in the number of trials, and within datasets there may be variation in the max number of trials in a session (for different ages or different sub-experiments), we set thresholds for the total trials rather than the fraction of useable trials.

Some datasets also exclude children for predominantly look at one side of the screen across trials ("side-bias"). Across our dataset, only 1.5% of administrations (contributing less than 1% of trials) looked at one side (right or left) more than 90% of the time across more than 75% of trials. Since this behavior is rare, excluding it or not probably won't make much of a difference. [I'm confused by this – side bias is really not equally distributed across datasets and it doesn't seem to always be a low trial/low data/trial issue. so maybe it matters for some datasets? by why?; are we sure it's not a hand-coding/data-processing/import error???]

Only including administrations that contribute at least 5 trials, results in 85.64 % (range:27.22% - 98.92%) of admins remaining.

These exclusions lead to an expected increase in ICC by 0 [-0.01, 0.01] and increase in test-retest reliability by 0.03 [-0.07, 0.12]. Differences in CDI correlation are not substantive (0.01 [-0.03, 0.05] for comprehension, and 0 [-0.03, 0.03] for production). CHECK TEXT

Excluding children for contributing too few trials is generally not necessary (Figure **??**). There is little difference in ICC reliability or CDI validity for 1-5 trials. For test-retest relability, the reliability is higher if there are at least a few trials from each test, although there are concerns that children who contribute fewer than 5 trials may be different in some way than children who contribute more, so the population generalizability may have issues.

It's acceptable, but probably not necessary, to exclude children for too few trials.

We also considered the effects of minimum trial requirements on RT data (Figure **??**). Only including administrations that contribute at least 5 trials, results in 60.28 % (range:5.56% - 100%) of admins remaining. TODO idk what a "reasonable" min RTs is – it might be lower

This is min of 5 trials! These exclusions lead to an expected increase in ICC by 0 [-0.06, 0.05] and increase in test-retest reliability by 0.09 [0.01, 0.17]. Differences in CDI correlation are not substantive (-0.02 [-0.17, 0.13] for comprehension, and 0 [-0.06, 0.07] for production). CHECK TEXT

For RT, it's reasonable to set low minimums for number of trials (like 2 trials) which does make for stronger correlations with CDI production. There are much stronger tradeoffs

between test-retest reliability and number of children included. It's also reasonable to not exclude.

Overall, exclusions based on amount of data per-trial or per-child come with trade-offs between amount of data and cleanliness of the data. Depending on the study design and goals, no exclusions, or light exclusions can both be reasonable choices.

Design recommendations

Our focus for this work was to explore how data processing choices affected the reliability and validity of the resultant measures. However, along the way, we came across a couple key experimental design questions that could impact reliability and validity.

Recommendation 5: [TODO discuss] Design studies to have a minimum of 10 trials per child, per condition, or as many trials as possible for individual-difference studies.

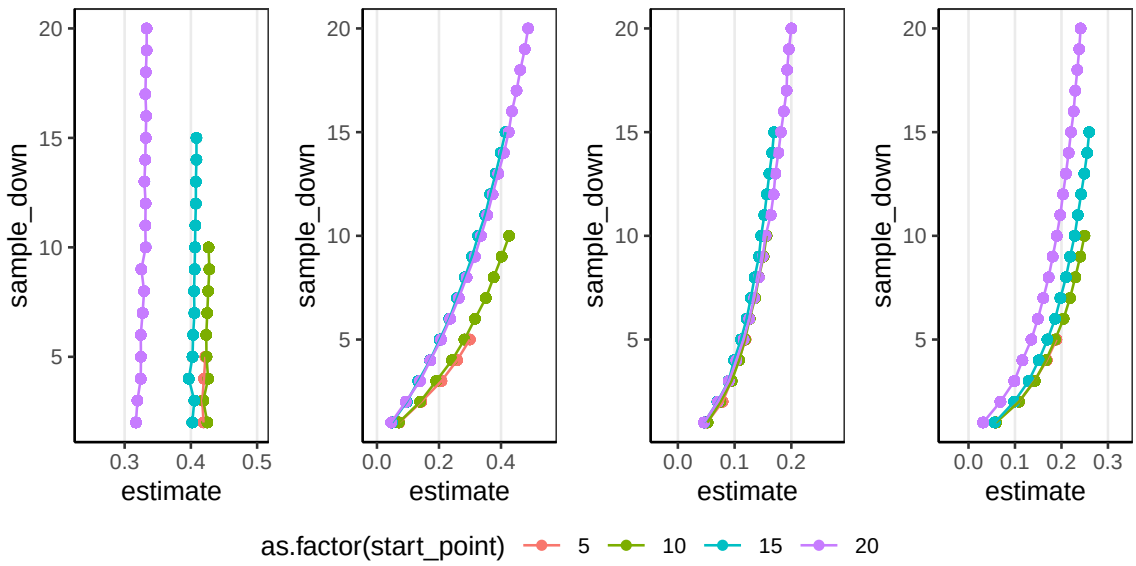


Figure 5. Simulations of measurement properties for different trial counts, created by downsampling trials for all children with different trial minimums. Shown for accuracy window of 400 to 3000ms with no trial- or kid- level exclusions.

Excluding children for low trial counts may increase test-retest reliability, but it also decreases the power and generalizability of results by excluding some children. Rather than retrospective exclusions, another approach is prospective, how many trials per child should one aim for to achieve adequate reliability and validity?

[TODO what is common practice for target number of trials / condition and for longitudinal v not?]

To address this prospective question, we considered how reliability and validity measures would have changed in we had fewer trials per child. (Ideally, we would also look at

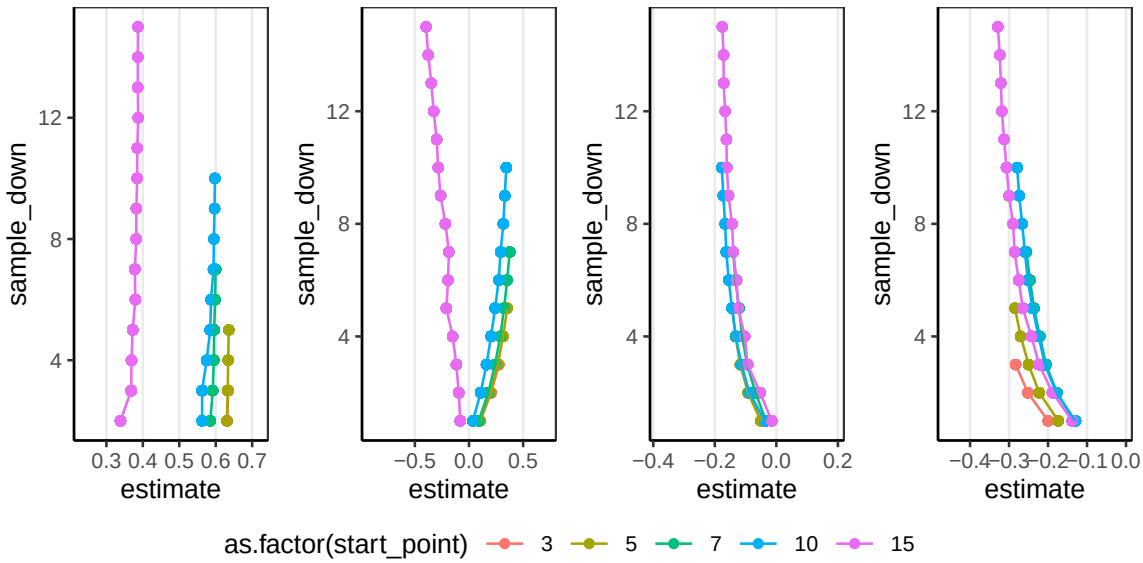


Figure 6. Simulations of measurement properties for different trial counts, created by downsampling trials for all children with different trial minimums. Shown for log landing RT with a minimum RT of 400ms.

more trials per child, but we don't know what these other trials would be, but it's easier to downsample the trials we do have to simulate fewer trials.) We filtered the data to include only children who had contributed at least N trials (for different values of N) and then repeatedly subsampled their trials and calculated our metrics on the subsets.

The results for accuracy are shown in Figure 5 and the results for RT are shown in Figure 6. ICC reliability is not sensitive to the number of trials per subject.

For accuracy, more trials per administration increases the test-retest reliability, up to 20 trials (the highest we measured). Validity with CDI measures also increased as trials increased, with smaller gains past 10 trials. Based on accuracy, we recommend at least 10 trials per condition in general, and as many as possible for individual differences studies where test-retest reliability is essential.

Nearly every trial a child does results in a useable accuracy measure, but only around 1/3 of trials result in a valid RT measure (in other cases, children start fixating the target or fixating neither).

For test-retest reliability, having at least 5 RT measures per administration is better than having fewer; gains beyond 5 are hard to measure because of low numbers of subjects with repeat administrations and enough trials, but 7 is likely better than 5. For CDI measures, there are substantial gains for having 2 trials compared to 1, and benefits to additional trials up to 7 trials. For individual differences in processing speed, as many trials as possible is very important (although we are limited by existing datasets to fully quantify the trade-offs).

Recommendation 6: Match target and distractor images based on animacy.

As mentioned in Recommendation 2, one concern with LWL trial design is that children may have preferences for which image to look at separate from the auditory stimuli, and some of their looking behavior may be independent of the auditory stimulus. One approach is to decrease the preference discrepancy between a pair of images, so that overall, in the absence of auditory cues, children prefer to look at both images equally. While there are a number of factors that could play into children's image preferences [citations], one factor that stood out in the data was animacy. Children have a strong preference for looking at animals and humans (and parts of humans). When trials have one animate and one inanimate, children look more at the animate in the pre-stimulus period (Figure 7) and are slower to respond when the inanimate is named. In contrast, for trials with target and distractor matched on animacy, children's early looking patterns are not sensitive to animacy. This difference is particularly relevant for the younger children, whose stimulus-contingent looking behavior is weakest, so more of the looking behavior is driven by animacy (Figure SXX).

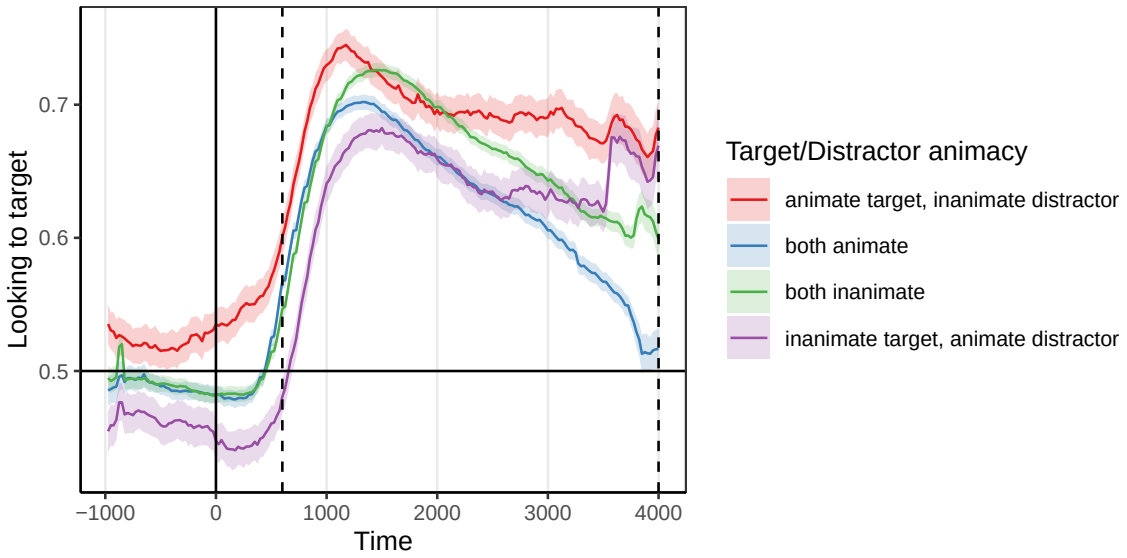


Figure 7. Looking time profiles for different combinations of target and distractor animacy.

Another potential concern regarding stimulus preferences is that repeating the same pairs of images may cause children to prefer to look at the image that was previously the target. We investigated this in a series of models (reported in Swhatever TODO), and found that effects of repeating target labels (and target and distractor images) are likely quite small, with unclear directionality. Repeating the same target nouns and target images, such as in a counterbalanced design, is an acceptable practice.

Animacy is not the only factor affecting children's viewing behavior, but it is an easy to measure one that makes a big difference. Other item variability could be accounted for in the statistical or analytic approach [citation for some approaches].

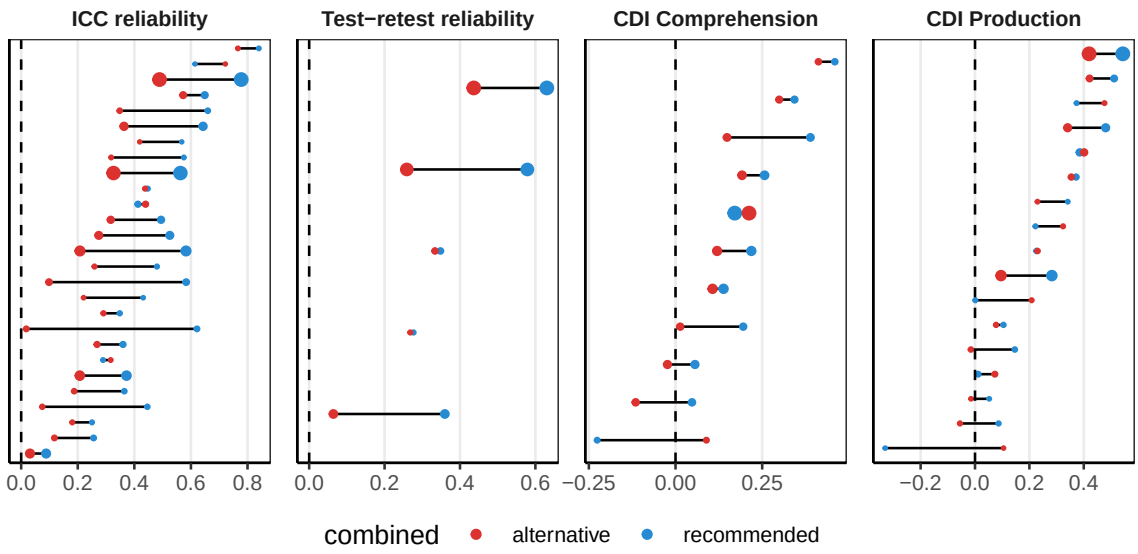


Figure 8. Placeholder for some graph about how much this can matter.

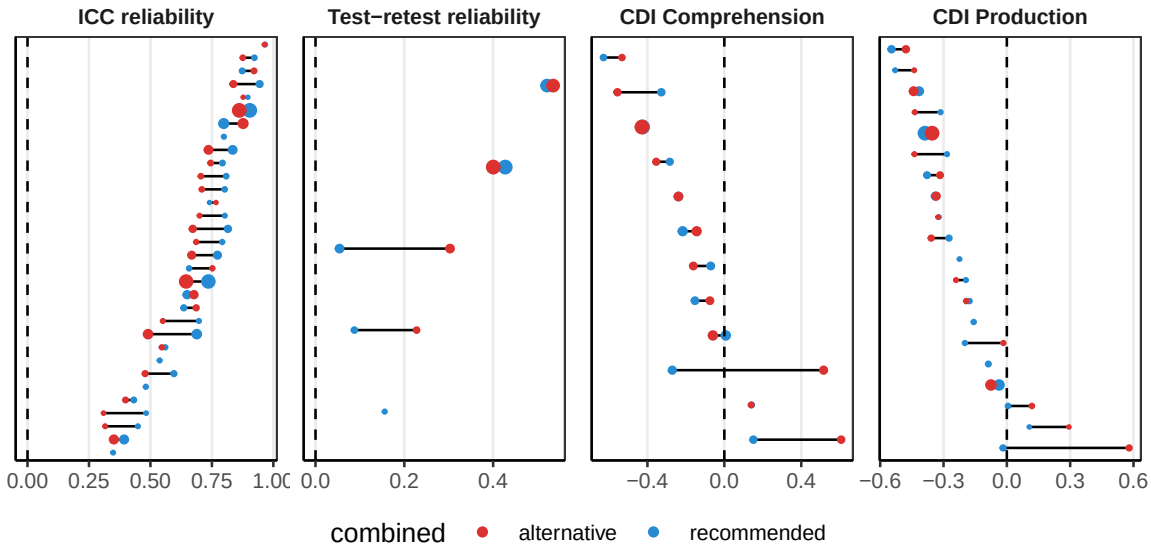


Figure 9. Placeholder for some graph about how much this can matter.

How much can this matter?

To quantify the potential cumulative consequences of different combinations of choices, we computed the difference in reliability when making all of the recommended choices with respect to data processing (in particular, X, Y, Z) to an instance in which researchers made all of the "worst-case" choices. [results - should this actually be in the results section?]

0.23 [0.18, 0.27] and increase test-retest reliability by 0.24 [0.15, 0.32]. Differences in CDI correlation are not substantive (0.05 [-0.02, 0.12] for comprehension, and 0.09 [0.03, 0.14] for production).CHECK NUMBERS!

and for RT 0.06 [0.03, 0.09] and increase test-retest reliability by -0.04 [-0.14, 0.06].

Differences in CDI correlation are not substantive (-0.07 [-0.21, 0.08] for comprehension, and -0.03 [-0.09, 0.03] for production).CHECK NUMBERS!

Discussion

Measuring infant looking time is notoriously noisy, making decisions about measurement practices highly consequential for the conclusions we draw about infant language development. How can we maximize reliable, valid signal when measuring infant word recognition? In our analyses, we took a practical, data-driven approach to this question, using a large-scale dataset of infant looking-while-listening data to quantify the impact of different analysis choices on core measurement properties, leading to six core recommendations for data processing and experimental design.

For data processing, our strongest take-homes from these analyses concern window choice and baseline correction. First, longer time windows (e.g., windows that end around 3000 ms post target word onset) maximize the reliability of accuracy measures (as measured by ICC), with no deleterious consequences for validity. While average accuracy tends to peak earlier in the timecourse of a trial (typically 1000-2000 ms post target word onset), leading shorter windows to be preferred in the literature, these findings suggest that later looking windows still capture signal that boosts the consistency of word recognition measures. Short accuracy windows might therefore be leaving reliable signal on the table. Second, baseline correction, while an intuitive solution to imbalances in the saliency of images, causes large decrements in reliability; we therefore generally recommend avoiding this practice, especially when maximizing reliability is a high priority (e.g., when investigating individual differences). Finally, we formulate recommendations for choosing reaction time cutoffs (around 400ms tends to maximize reliability) and find that implementing trial- and child-level exclusions based on the amount of data contributed tend to have minimal consequences. Overall, choices about windows of analysis, approaches to exclusions, and how scores are computed can all have meaningful impacts on measure reliability.

[recommendations around design - hype the animacy finding]

could mention a meta-point that looking only at grand-average time courses may not give the best intuitions about child or item level analyses?

Limitations & Generalizability

The scope of our recommendations is fundamentally limited by the characteristics of the Peekbank dataset. First, while the Peekbank database is by far the largest dataset of its kind, and one of the largest harmonized infant eyetracking datasets, it is not a comprehensive or representative set of looking-while-listening studies. To highlight just a few of these limitations, (1) the data is predominantly English word recognition collected from infants living in the U.S., and (2) the dataset is a convenience sample from labs who made their data openly available or otherwise accessible on request from the Peekbank team. Looking-while-listening studies as a whole are likely more heterogeneous than what is represented in Peekbank — the datasets considered here come predominantly from a small

group of labs or networks of researchers with similar training in looking-while-listening data collection.

At the same time, there is also substantial heterogeneity in the studies included in Peekbank, due to many studies pursuing different research questions and therefore making a variety of distinct design choices (e.g., surrounding target words, image properties, and characteristics of the auditory stimuli). To mitigate this concern, our analyses focused exclusively on what we term "vanilla" trials, i.e., trials that present familiar words with familiar images with limited or no targeted manipulations of the stimuli. The main benefit of focusing on vanilla trials is that it allows for a clear interpretation of reliability and validity — trials are truly designed to measure word recognition, rather than some other aspect of language processing. We view the remaining heterogeneity of study properties (e.g., specific words; differences in age) as a strength of our approach — our findings generalize broadly across a wide range of items and ages. However, by focusing on vanilla trials only, we also limit our conclusions to measuring familiar word recognition; our findings do not address how data processing and design practices would generalize to other estimation goals, e.g., estimating the magnitude of differences between two experimental conditions (though there is good reason to think that some of the more consequential choices, e.g. baseline correction, are likely have a similar impact).

While we summarize the choices that typically maximize reliability, we note that we often see a range of choices with similar outcomes in our analyses. For example, in determining the best time window for computing accuracy, a choice of ending a time window at 2800 vs. 3000 or 3000 ms vs. 3200 ms leads to very similar outcomes. There is no one choice of time window that we can definitively infer as the "best" for maximizing reliability — instead, we see a pattern of graded improvements in reliability that follow a predictable pattern (longer time windows tend to increase reliability holding the onset of the accuracy window constant). While we sometimes articulate which choices lead to the highest reliability descriptively (e.g., time windows from 300-4000ms [check once finalized] leads to the highest reliability values for accuracy), this is not intended as the definitive window choice. We offer these windows as data-driven default choices (especially in cases where the decision would otherwise be ad-hoc), but from our perspective, the important takeaways concern the factors that tend to have a consistent and generalizable impact on reliability (e.g., amount of data per trial; difference scores vs. raw scores).

We focused our explorations on the questions that are specific to LWL and where data-driven results based on a combined dataset can be most helpful. Other guidance both on LWL specific design considerations and options for coding LWL data are discussed elsewhere ([fernald book chapter, e.g.]). More general advice on experiment design and analysis, especially of developemental datasets, is discussed elsewhere as well [TODO].

Conclusions

Experimental design and data processing decisions can have profound impacts on the properties of infant measures. Our findings show how compiling large-scale datasets such as Peekbank can help us estimate the impact of a wide variety of experimenter decisions,

generalizable across a wide range of ages, stimuli, and experimental contexts. Our findings are summarized in a short list of recommendations that should help researchers aiming to maximize the reliability of infant word recognition measure: [summarize]. Establishing consistent practices and principles for reliable and valid measurement of infant looking sets the stage for asking central questions about how language skill develops.

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